

Print Server Administration & User Guide

Internal IT Knowledge Base / SOP Manual

Document Type	Internal IT Knowledge Base / User Guide / SOP
Audience	IT Administrators, IT Support, Help Desk, and Non-IT Staff
Environment	Windows Server
Domain	lab.local
Primary Service	PRINT01 – Print Services
Example Printer	LAB-Printer01
Related Services	DC01- Active Directory Domain Services (AD DS), DNS, DHCP01 and CLIENTPC

1. Purpose

This document provides a standardized procedure for installing, configuring, managing, sharing, troubleshooting, and deploying network printers using a Windows Server Print Server.

The goal is to provide centralized printer management so that IT administrators can manage printers and printer drivers from a single server while allowing authorized users and workstations to access shared printers.

2. Scope

This procedure applies to:

- IT Support personnel
- System Administrators
- Employees using company-managed computers
- Domain-joined Windows workstations
- Windows Server-based print services
- Shared network printers

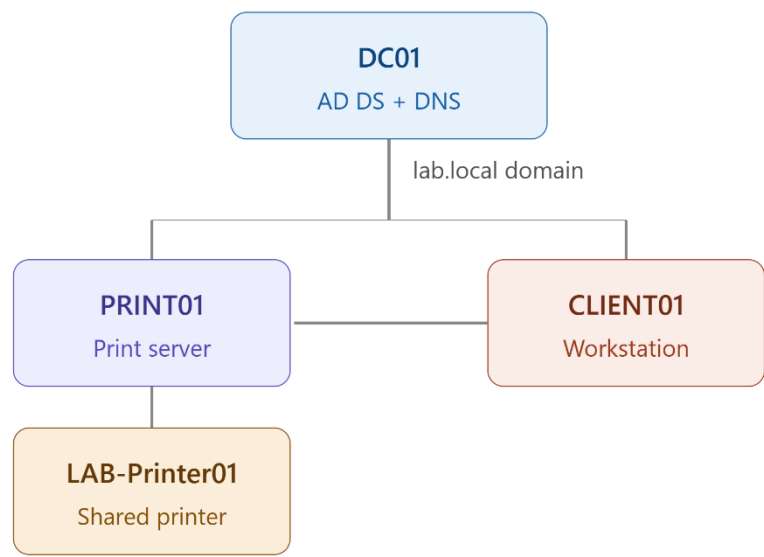
This guide covers:

- Print Server installation
- Printer driver installation
- Printer creation
- Printer sharing
- Printer permissions
- Client printer connection
- Group Policy printer deployment
- Printer testing

- Basic troubleshooting
- Printer administration best practices

3. Environment Overview

The example environment uses the following infrastructure:



Example Configuration

Component	Configuration
Domain	lab.local
Domain Controller	DC01
Print Server	PRINT01
Example IP	192.168.1.104
DNS Server	DNS Server
Client	CLIENT01
Printer	LAB-Printer01
Printer Share	LAB-Printer01
Printer Path	\\PRINT01\LAB-Printer01

4. Roles and Responsibilities

IT Support

IT Support is responsible for:

- Installing approved printer drivers
- Adding and removing printers
- Troubleshooting printer connectivity
- Managing printer queues
- Verifying printer permissions
- Assisting users with printer installation
- Escalating hardware problems when necessary

System Administrator

System Administrators are responsible for:

- Managing the Print Server
- Managing printer drivers
- Configuring printer sharing
- Configuring Group Policy deployment
- Managing printer security
- Maintaining server availability
- Reviewing print-server configuration

End Users

End users are responsible for:

- Using assigned printers appropriately
- Reporting printer problems to IT
- Checking for basic issues before contacting IT
- Avoiding unauthorized printer-driver installation
- Reporting printer errors accurately

5. Prerequisites

Before configuring the Print Server, verify the following.

Server Requirements

- Windows Server installed
- Server has a static IP address
- Server has a valid computer name
- Server is joined to the domain
- DNS points to the domain DNS server
- Network connectivity to clients is working
- Appropriate administrative permissions are available

Example:

Server Name: PRINT01
Domain: lab.local
DNS Server: DC01

6. Install the Print Server Role

Log in to PRINT01 using an administrative account.

Open:

Server Manager

Select:

Manage → Add Roles and Features

Select:

Role-based or feature-based installation

Select:

PRINT01

Under Server Roles, select:

Print and Document Services

When prompted, select:

Add Features

Continue through the wizard.

Under Role Services, select:

Print Server

Click:

Next → Install

Wait for the installation to complete.

7. Open Print Management

After installing the Print Server role:

Open:

Server Manager → Tools → Print Management

The management console should display:

```
Print Management
├── Print Servers
│   └── PRINT01
│       ├── Drivers
│       ├── Forms
│       ├── Ports
│       └── Printers
```

Print Management is the primary administration console for managing printers and printer drivers.

8. Install a Printer Driver

A printer driver allows Windows to communicate with a printer.

When adding a printer, Windows may provide existing drivers.

For a lab environment, an available compatible driver such as:

Microsoft PCL6 Class Driver 2

can be used for testing.

Important

Important: A printer driver is not the physical printer. The driver only provides the software interface required for Windows to communicate with the printer.

For production environments, use the manufacturer's approved driver whenever possible.

Examples:

```
HP → HP-approved driver
Canon → Canon-approved driver
Brother → Brother-approved driver
Epson → Epson-approved driver
```

9. Add a Printer

Open:

Print Management → Print Servers → PRINT01 → Printers

Right-click:

Printers → Add Printer

Select the appropriate printer connection method.

For a real network printer, the recommended configuration is generally a TCP/IP printer connection.

Example:

Printer IP Address:
192.168.1.200

For the driver, select the appropriate manufacturer or compatible driver.

For a lab environment where no physical printer is available, a compatible generic driver may be used for practicing Print Server administration.

10. Configure Printer Name

Use a standardized naming convention.

Example:

LAB-Printer01

In a production environment, a more descriptive naming convention may be used.

Example:

MANILA-FLOOR01-PRINTER01

or:

FINANCE-PRINTER01

The name should make it easy for IT Support and users to identify the printer.

11. Share the Printer

Open:

Printer Properties → Sharing

Enable:

Share this printer

Specify the share name:

LAB-Printer01

The resulting network path is:

\\PRINT01\LAB-Printer01

Users can connect to the printer using this path.

12. Printer Permissions

Printer permissions control which users can use or manage the printer.

Open:

Printer Properties → Security

A typical configuration might include:

Group	Permission
Domain Users	Print
IT Support	Print + Manage Documents
Print Administrators	Manage Printer + Manage Documents
Domain Admins	Full administrative control

Recommended principle

Users should receive only the permissions necessary to perform their job.

For normal employees:

Print

is generally sufficient.

IT administrators may require:

Manage this printer
Manage documents

13. Test the Printer

After creating the printer, perform a test print.

On PRINT01:

Printer Properties → General → Print Test Page

Verify that:

- The printer accepts the job
- The print queue works
- The correct driver is being used
- No errors are displayed

Open:

See What's Printing

to monitor the print queue.

14. Connect a Client Computer

On a domain-joined workstation such as CLIENT01:

Press:

Windows + R

Enter:

\\PRINT01

Press Enter.

The shared printer should be visible.

Double-click:

LAB-Printer01

Windows will attempt to connect to the shared printer.

The printer should then appear under:

Settings → Printers & scanners

15. Test from the Client

On CLIENT01:

Go to:

Settings → Printers & scanners

Select:

LAB-Printer01

Select:

Printer properties

Click:

Print Test Page

Verify that the print job reaches PRINT01.

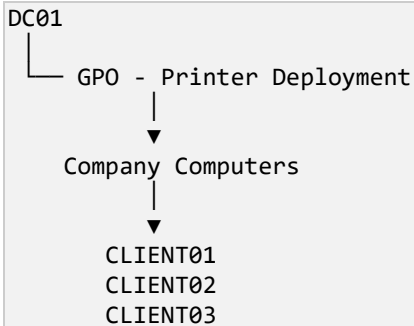
On the Print Server, check:

Print Management → Printers → LAB-Printer01 → See What's Printing

16. Deploy the Printer Using Group Policy

For an organization with many computers, manually installing printers on every workstation is inefficient. Group Policy can automatically deploy printers to domain computers.

Example:



Create a GPO:

GPO - Printer Deployment

Link it to the appropriate workstation OU.

For example:

```

lab.local
├── Company Computers
│   └── Workstations
  
```

The printer should use the shared path:

\\PRINT01\LAB-Printer01

17. Update Group Policy

On the client computer, open Command Prompt as Administrator.

Run:

gpupdate /force

If necessary, restart the workstation:

shutdown /r /t 0

After restarting, verify the applied policies:

gpresult /r

Look under:

COMPUTER SETTINGS

Verify that:

GPO - Printer Deployment

appears under the applied GPOs.

18. Verify Printer Deployment

After Group Policy has been processed:

Go to:

Settings → Printers & scanners

Verify that:

LAB-Printer01

is installed.

You can also check:

Control Panel
→ Devices and Printers

The printer should be available to the workstation.

19. Printer Queue Management

The print queue contains jobs waiting to be processed.

On PRINT01:

Print Management → Printers → LAB-Printer01

Right-click:

Open / See What's Printing

IT Support can:

- Pause a job
- Resume a job
- Cancel a job
- Restart a job
- Clear stuck jobs

Example

If a user reports:

User Report: “My document has been stuck printing.”

IT Support should check the queue before restarting services or reinstalling the printer.

20. Common Printer Problems

Problem 1 — Printer does not appear

Check:

```
Client is connected to the domain/network.  
PRINT01 is online.  
The printer is shared.  
The client can access:  
\\PRINT01  
The correct GPO is applied.
```

Run:

```
gpresult /r
```

Problem 2 — Cannot connect to the printer

Test:

```
ping PRINT01
```

Then:

```
\\PRINT01
```

If \\PRINT01 cannot be opened, investigate:

- DNS
- Network connectivity
- Windows Firewall
- Server availability
- SMB connectivity
- Permissions

21. Problem — Print Job Gets Stuck

On PRINT01:

Open:

```
Print Management → LAB-Printer01 → See What's Printing
```

Check the queue.

If necessary:

- Cancel the stuck job.
- Retry the print job.
- Check the printer status.
- Verify the printer driver.
- Restart the Print Spooler service if required.

PowerShell:

```
Get-Service Spooler
```

Restart:

```
Restart-Service Spooler
```

22. Problem — Printer Spooler Is Not Running

Check:

```
Get-Service Spooler
```

If stopped:

```
Start-Service Spooler
```

To restart:

```
Restart-Service Spooler
```

23. Problem — Client Cannot Find PRINT01

Check DNS:

```
nslookup PRINT01
```

The name should resolve to the correct IP address.

Also test:

```
ping PRINT01
```

If DNS does not resolve correctly, investigate the client's DNS configuration.

For an Active Directory environment, domain clients should normally use the organization's internal DNS server.

24. Problem — Printer Works on Server but Not Client

Check the following:

1. Printer is shared
2. Share name is correct
3. Client can access \\PRINT01
4. User has Print permission
5. Printer driver is compatible
6. GPO is applied
7. Print Spooler is running

25. Important: Microsoft Print to PDF

Important: Microsoft Print to PDF should not be treated as a normal shared network printer for this lab.

It is a Windows virtual printer rather than a physical/network printer designed for centralized Print Server sharing.

For a Print Server lab, use a printer/driver configuration that supports shared printer deployment.

For production environments, use the printer manufacturer's supported driver.

26. Security Best Practices

IT administrators should:

- Use approved printer drivers.
- Avoid downloading unknown drivers.
- Restrict printer administration to IT personnel.
- Give normal users only Print permission.
- Keep the Print Server patched.
- Monitor printer availability.
- Remove unused printers.
- Remove outdated drivers when appropriate.
- Use standardized printer naming.
- Document printer IP addresses.
- Document physical printer locations.
- Use Group Policy for centralized deployment where appropriate.

27. IT Troubleshooting Checklist

When a user reports a printer problem:

- ☐ Confirm the affected printer name.
- ☐ Confirm the user's computer name.
- ☐ Confirm the user can access the network.
- ☐ Check whether PRINT01 is online.
- ☐ Check whether the printer is shared.
- ☐ Test \\PRINT01.
- ☐ Check printer permissions.
- ☐ Check the printer queue.
- ☐ Check the Print Spooler service.
- ☐ Verify the printer driver.
- ☐ Run gpupdate /force if GPO deployment is involved.
- ☐ Run gpresult /r to verify Group Policy.
- ☐ Test printing from another workstation.
- ☐ Document the resolution.

28. End-User Quick Guide

How to connect to a company printer

If IT has provided a printer path:

Press:

```
Windows + R
```

Enter:

```
\\PRINT01
```

Press Enter.

Find the appropriate printer and double-click it.

Alternatively, IT may automatically install the printer through Group Policy.

How to print a document

1. Open the document.
2. Select File → Print.
3. Select the appropriate company printer.
4. Confirm the number of copies.
5. Click Print.

29. When to Contact IT Support

Contact IT if:

- The printer is missing.
- The printer shows Offline.
- Documents remain stuck in the queue.
- You receive a printer-driver error.
- You cannot connect to the printer.
- The printer repeatedly fails to print.
- The printer displays a hardware error.
- You are unsure which printer to use.

When reporting an issue, provide:

```
Computer:
CLIENT01

Printer:
LAB-Printer01

Problem:
Unable to print

Error:
<error message>
```

When it started:
<date/time>

Providing this information allows IT Support to troubleshoot the issue faster.

30. Administrative Reference

Server	PRINT01
Domain	lab.local
Shared Printer	LAB-Printer01
Printer UNC Path	\\PRINT01\LAB-Printer01
Print Management	Server Manager → Tools → Print Management
Print Spooler	Get-Service Spooler
Restart Print Spooler	Restart-Service Spooler
Force Group Policy Update	gpupdate /force
Check Applied Group Policy	gpresult /r
Test DNS	nslookup PRINT01
Test Connectivity	ping PRINT01

31. Change Management

Any production printer configuration change should be documented.

Examples of changes requiring documentation:

- New printer
- Printer removal
- Printer IP change
- Driver replacement
- Share-name change
- Permission change
- GPO deployment change
- Printer relocation

Example change record:

Change:
Added Finance Department printer

Printer:
FINANCE-PRINTER01

Server:
PRINT01

Driver:
<approved driver>

Share:
\\PRINT01\FINANCE-PRINTER01

GPO:
GPO - Finance Printer Deployment

Date:
<date>

Performed By:
<IT administrator>

Status:
Completed

32. Final Architecture

Your completed homelab should eventually look like:

